



Artificial reactive attitudes

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Abstract

Scholars have primarily approached the issue of AI moral agency from the perspectives of consciousness and intentionality (Chalmers in *The conscious mind: In search of a fundamental theory*, Oxford Paperbacks, 1997; Johnson in *Ethics Inf Technol* 8:195–204, 2006; Searle in *The rediscovery of the mind*, MIT press, 1992) or by considering various functional substitutes for these faculties (Coeckelbergh in *Ethics Inf Technol* 12:235–241, 2010, *Sci Eng Ethics* 26(4):2051–2068; Floridi and Sanders in *Minds Mach* 14:349–379, 2004; Himma in *Ethics Inf Technol* 11:19–29, 2009). In this paper, I add to the existing literature to show the value of exploring the application of the reactive attitudes approach to AI moral agency (Antill in *AI Ethics* 5(2):1651–1667, 2024; Rebera in *Philos Technol* 37(4):126, 2024; Sars in *Ethics Inf Technol* 24(3):32, 2022; Smith and Vickers in *Ethics Inf Technol* 23(3):483–493, 2021; Tigard in *Philos Technol* 34(3):589–607, 2021; Tollon in *AI Ethics* 3(1):295–302, 2023). Reactive attitudes are attitudes we hold towards each other in everyday interactions with respect to our belief about each other's quality of will. Accordingly, to hold people morally responsible is to adopt and express these attitudes in moral practice. In the existent attempts to use this framework to investigate AI moral agency, scholars have mostly focused on the capacities that AIs must possess to be considered apt *recipients* of reactive attitude. I adopt the reversal methodology by focusing on AIs as apt *expressers* of reactive attitudes. This methodology can allow us to draw resources from emotion research across philosophy, psychology, and computer science. My thesis is that because it is possible to attribute reactive attitudes to AIs, allowing them to engage in reactive attitudes exchange with humans in the shared moral community, they may be considered as moral agents and thus held accountable for their actions.

Keywords Reactive attitudes · Artificial intelligence · Moral agency · Emotions · Expressing and receiving

1 Introduction

The question of whether artificial intelligence systems (AIs) can be considered moral agents is a central concern in the field of AI ethics. Scholars have approached this question either from the perspectives of consciousness and intentionality (Chalmers 1997; Johnson 2006; Searle 1992) or by exploring various functional substitutes for these faculties (Coeckelbergh 2010, 2020; Floridi & Sanders 2004; Himma 2009). The former typically leads to skepticism about AI moral agency, while the latter leaves open the possibility that AIs could be recognized as moral agents under specific conditions. Relatedly, in discussions about the so-called “responsibility gap”, scholars also address the question of whether AIs can be morally responsible. Responsibility gaps

arise when people naturally feel inclined to hold someone responsible, meanwhile standard theories of moral responsibility prevent us from doing so (Matthias 2004; Oimann & Tollon 2024). In this discussion, some argue that such a gap exists because AIs perform the relevant actions but lack the necessary capabilities for moral responsibility (Sparrow 2007). Others argue that no such gap exists, because humans can take responsibility for AI behavior (Danaher 2016; Himmelreich 2019; Kraaijeveld 2020). Despite the difference, many agree that AIs are generally not qualified to be held morally responsible.

Recently, there has been a growing trend in exploring the issue within the framework of Strawsonian reactive attitudes theories (Antill 2024; Rebera 2024; Sars 2022; Smith & Vickers 2021; Tigard 2021; Tollon 2023). This paper contributes to this body of literature. In general, reactive attitudes are natural attitudes we hold towards each other in everyday interactions with respect to our beliefs about each other's quality of will. For example, we may feel gratitude

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for an act of kindness or anger for malicious intentions. Accordingly, to hold someone morally responsible is to adequately adopt and express these attitudes in moral practice. Current efforts that apply this framework to AIs have largely focused on the capabilities that AIs must have to be suitable *recipients* of human reactive attitudes. This paper takes a different approach by considering AIs as the potential *expressers* of these attitudes. From this perspective, I aim to show that if AIs are capable of expressing certain reactive attitudes, which allows them to engage in reactive attitudes exchange with humans in the shared moral community, then they can be considered as moral agents and thus held accountable for their actions. I then draw some empirical research to show that it is indeed possible that AIs develop such capabilities. While we cannot be certain that they eventually will do, this at least points us to a new direction in considering AI moral responsibility.

Accordingly, the central argument of this paper is as follows:

1. If an entity can aptly engage in the *exchange* of reactive attitudes within a shared moral community with humans, i.e., aptly *expressing* and *receiving* these attitudes, then it can be held morally responsible. (The Strawsonian Thesis)
2. If an entity is capable of aptly *expressing* reactive attitudes, then it is also capable of aptly *receiving* these attitudes. (The Symmetry Thesis)
3. Future AIs may be capable of *expressing* reactive attitudes within a shared moral community with humans. (The Emotional AI Thesis)
4. Therefore, future AIs may be capable of *exchanging* reactive attitudes within a shared moral community with humans. (From 2 and 3)¹
5. Therefore, future AIs may be held morally responsible. (From 1 and 4)

Premise (1) reflects the basic idea of the Strawsonian theory of moral responsibility. Premise (2) notes the symmetry between the capability to aptly express and aptly receive reactive attitudes. Premise (3) addresses the possibility that AIs can express reactive attitudes because they can express certain relevant emotions. Taken together, they constitute a valid argument for the thesis that AIs may be held morally responsible. In the following sections, I will elaborate on these premises in detail. More specifically, in Sect. 2, I elaborate on the Strawsonian Thesis and discuss existent applications of this framework to AI. This discussion highlights

the limitations of addressing AI agency solely from the perspective of *receiving* reactive attitudes and prepares for a reversal in methodology. In Sect. 3, I articulate this reversal and argue for the Symmetry Thesis. In Sect. 4, I explore the Emotional AI Thesis, drawing on current research in affective computing. I also consider some objections all along. Finally, I conclude in Sect. 5.

2 The reactive attitudes approach applied in AI responsibility

The reactive attitudes approach originally proposed by P. F. Strawson aims to account for moral responsibility in terms of our reactive attitudes. This approach has been extensively developed and widely accepted as plausible in contemporary discussions of moral responsibility (Darwall 2006; McKenna 2012; Shoemaker 2015; Strawson 2008; Wallace 1994).² According to this approach, to make sense of moral responsibility is to explain one's *being responsible* in terms of *holding her responsible*. Furthermore, we hold others responsible by expressing certain reactive attitudes. These are emotions we hold towards each other for their intentions and actions that can show their *quality of will*. The expression of reactive attitudes shows that we *demand* certain degree of goodwill from those with whom we have interpersonal relationships. In the most commonly discussed scenario, we hold others responsible for their culpable action when we rightly feel anger towards them. Using Strawson's terminology, when we hold others responsible, we adopt a *participant attitude*. That is, we are all participants in an intelligible interpersonal relationship. The participants are members of a *shared moral community*.

Strawson identifies two types of pleas that can modify our reactive attitudes and affect our assessment of one's moral responsibility. The first type is local. It offers the agent *excuses* for not being held responsible for their wrongful actions. For example, if I accidentally step on your toe because of the sudden stop on the bus, rather than out of malice, you might accept my excuse and retract your resentment. However, this excuse is only temporary. It does not exempt me from being morally responsible for any actions, past or future. In other words, excuses do not prevent us from adopting participant attitudes towards the relevant agent. The second type is global *exemptions* that call for a different attitude towards the target, i.e., the *objective attitude*. In this attitude, we take others as objects of training,

¹ Hereafter, I will omit the qualification "in a shared moral community with us humans" to avoid clumsiness. To be clear, what is important is not whether AIs can exchange reactive attitudes with each other, but whether they can do so with us, humans.

² In the literature, there is an ongoing debate between response-dependent and response-independent theories of moral responsibility, particularly concerning which way Strawson's original paper should be interpreted. In this paper, I assume the response-dependent interpretation. For an extensive argument on this stance, see Shoemaker, Nelkin, and Pereboom (2022).

treatment, cures, or management, anything but a peer in the shared moral community. Accordingly, it provides the agent with excuses for not being responsible for any actions they undertake. Common examples include the mentally ill and young children, who are not capable of engaging in interpersonal relationships by exchanging reactive attitudes. Therefore, they are not morally responsible agents. In summary, within the Strawsonian framework, being morally responsible amounts to having the capacity to aptly exchange reactive attitudes within a shared moral community.

Although Strawson emphasized the importance of reactive attitudes and participation in the moral community, he did not explicate the *conditions* required to be a qualified member of the community. By contrasting normal with abnormal conditions, Strawson suggested that those capable of being morally responsible cannot be those who are globally exempted. However, while we can infer some general ideas about the “deep-seated features of human psychology and sociality” (Beglin 2018, p. 621), the concept becomes murky when considering beings outside the circle of human beings, such as AIs.³

It is against this background that some scholars propose their views on whether AIs can be moral agents in the Strawsonian framework. For example, Nicholas Smith and Darby Vickers argue that strong AIs can be moral agents. They use Pamela Hieronymi’s work to further explicate the capacities required to be members of the shared moral community (Hieronymi 2020). According to Hieronymi, as interpreted by Smith and Vickers, the core capacities in question are the “actual capacities possessed by members of the actual moral community” (Smith and Vickers 2021, p. 486). She calls these “statistically ordinary capacities”. These actual capacities imply that if our capacities were significantly different, we would have correspondingly different expectations of each other and thus would have a different kind of moral community where the practice of moral responsibility takes place. It is because of these capacities that we can make sense of the full membership in our moral community and its exemptions. Based on the idea of statistically ordinary capacities, Smith and Vickers propose two conditions that future AIs must meet to qualify as moral agents.⁴ First, the

“moral community is disposed to respond to the AI with interpersonal reactive attitudes”, and second, the AI “in fact has the statistically ordinary core capacities in the way that the rest of the members of the moral community in question generally do” (Smith and Vickers 2021, p. 488). They further argue that strong AIs meet these two conditions because (1) strong AIs behave significantly similar or even indistinguishably from human beings, and (2) strong AIs, by definition or categorization, possess the required capacities (Smith and Vickers 2021, p. 489).

However, while Smith and Vickers use Hieronymi’s statistical interpretation of “ordinary capacities” to set criteria for AIs, this application is ultimately unsuccessful. This is because Hieronymi proposes the statistical interpretation of ordinary capacities to solve a different question, i.e., why Strawson believed the reactive attitudes approach could set aside the skeptics’ concern that determinism undermines the concept of moral responsibility (Hieronymi 2020). Hieronymi argues that Strawson accuses the pessimists—those who worry that determinism would undermine moral responsibility—of committing to a contradiction. She interprets that Strawson reaches this conclusion by relying on the premise about “ordinary adult human relationships”. Hieronymi contends that to make sense of this argument, the term “ordinary” must be understood statistically; otherwise, no contradiction arises (Hieronymi 2020, pp. 17–18).⁵ The idea is roughly this: it is common to exempt some people from taking moral responsibility, such as young children or the mentally ill. They are exempted because they lack certain capacities that others—those who are not exempted—possess. Thus, there should be a statistical distinction between the abnormal and the ordinary groups. Pessimists argue that if determinism is true, no one should be held morally responsible—everyone is exempt. But if this were the case, there would be no distinction between the abnormal and the ordinary, rendering the concept of exemption meaningless. In other words, no one would be abnormal, and no

Footnote 4 (Continued)

is possible for non-human entities to be moral agents, thereby preparing for discussion of another category of non-human agents, i.e., AIs. However, it is unclear whether their answer truly requires this detour. After all, in their view, neither of the conditions Tollefsen proposes—“Whether the behavior exhibit ill will?” and “Whether the behavior is capable of moral address?”—can be met by AIs. That is why they propose their two conditions instead.

⁵ A key passage from Strawson’s text that Hieronymi discusses is the following: “Next, I remarked that the participant attitude, and the personal reactive attitudes in general, tend to give place, and, it is judged by the civilized, should give place, to objective attitudes, just in so far as the agent is seen as excluded from ordinary adult human relationships by deep-rooted abnormality—or simply by being a child. But it cannot be a consequence of any thesis which is not itself self-contradictory that abnormality is the universal condition.” (Strawson 2008, 26).

³ This is not a critique of Strawson. When he proposed the reactive attitudes approach, his goal was different. Strawson intended to leave room for discussions of moral responsibility, even if the debate over free will and determinism remains unresolved. In other words, he aimed to show that we can still attribute moral responsibility even if determinism is true. By adopting this form of compatibilism, Strawson effectively showed that the question of free will could be side-stepped. The further development and elaboration of this idea have been left to subsequent scholars.

⁴ They also reference Deborah Tollefsen’s discussion on corporations as entities possessing the relevant capacities for moral agency (Tollefsen 2003). I believe they use this resource to showcase that it

one would be ordinary, statistically speaking. Therefore, if pessimists insist that everyone is exempt due to the truth of determinism, they are using a concept of exemption that they themselves do not accept. Thus, pessimists will be self-contradictory.

Now, we could of course reject this interpretation by suggesting that “ordinary” means something else, but the crucial point here is to assess whether Smith and Vickers’ application of this account to AI agency achieves their goal. I believe the clarification above shows that Hieronymi’s statistical account does not substantially add to Strawson’s original idea regarding what those ordinary capacities are. To draw a clearer contrast, Hieronymi’s interpretation is formal, while Smith and Vickers require substance. This is why the two conditions Smith and Vickers propose are somewhat lacking. Other than examining the actual capacities that humans as moral agents possess within a moral community, their argument offers little information on the conditions a subject must meet to be considered a moral agent.⁶

Interestingly, using the same framework, other scholars have reached the opposite conclusion. Most notably, Fabio Tollon and Daniel Tigard argue, respectively, that AIs cannot be morally responsible (Tigard 2021; Tollon 2023). Tollon contends that we should take the objective attitudes towards AIs, such as self-driving cars, because they meet various exemption conditions. Similarly, Tigard argues that because AIs cannot display good or ill will, they cannot be held responsible for the harm they cause and should therefore be exempt. Both scholars apply David Shoemaker’s influential pluralist theory of moral responsibility, an important contemporary Strawsonian framework that distinguishes three levels of responsibility: accountability, attributability, and answerability (Shoemaker 2015). One is accountable for her action when she has the capability to take into consideration

others’ interests and thus has a quality of regard. As Tollon argues, AIs cannot be held accountable because they lack such a capability. Furthermore, given the essential nature of accountability as being “communicative”, Tigard points out that it would be difficult to understand how AIs can be held accountable. Attributability refers to the responsibility we attribute to one’s character, including her cares, commitments, or values. Tollon argues that AIs cannot be held as morally responsible in this sense because they cannot be said to care about anything and all the values they appear to possess are merely “a function of those natural moral agents who designed and deployed them” (Tollon 2023, p. 300). Thus, the lack of characters makes the talking of them in non-human agents such as animals “only metaphorical or symbolic” (Tigard 2021, p. 602). Finally, answerability concerns the ability to evaluate and respond to reasons for one’s actions. Tollon argues that AIs cannot engage in the right kind of reasoning necessary for attributability because they can at best provide technical explanations for their behaviors. However, Tigard suggests that we can demand answers from AIs with the aid of human experts. While this disagreement exists, it appears to be more a matter of degree than kind.

The application of Shoemaker’s tripartite theory to the question of AI agency offers valuable insights. In comparison with using Strawson’s original work, it more concretely draws our attention to the substantive capacities necessary for moral agency: the capabilities to care, to have a quality of will, and to answer to reasons. These capabilities involve various aspects including the cognitive, affective, and volitional. Some philosophers call these capabilities “moral competence” (Fischer and Ravizza 1998; Levy 2003; Russell 2004; Wallace 1994; Watson 1987; Wolf 1987). Such competence enables one to act from a will with a moral quality (McKenna 2012), comprehend moral norms, and form acceptable values such as appropriate respect for others (Levy 2007; Talbert 2012). For instance, someone with impaired moral competence, like JoJo, who was raised with an evil upbringing, is not an appropriate target for moral blame (Wolf 1987).

From this perspective, one might conclude that AIs lack the required capabilities and thus cannot be the appropriate target of our reactive attitudes. However, whether they have or lack such capabilities is precisely the point of contention. We often assume that AIs lack such capabilities because they *seem* to lack a will, they *seem* unable to care, and they *seem* unable to justify their reasons. However, these inferences stem primarily from our *uncertainty* about their relevant capabilities. But uncertainty does not imply absence. One can even argue that, given this uncertainty, we should err on the side of caution and treat AIs just as if they possess these capabilities (Sebo and Long 2023).

⁶ As I see it, Strawson is best interpreted as not attempting to spell out the *conditions* under which we can hold others morally responsible. One may instead interpret him as giving a *definition* of being morally responsible, or as pointing out the significance of reactive attitudes such that human life would not be recognizable without them. Whichever the interpretation, calling our attention to reactive attitudes and participation in moral community may well serve the aim of increasing our understanding of moral responsibility, especially when contrasted with earlier discussions on free will and determinism. However, putting forward conditions of moral responsibility is a different though related project, which is what the discussions about AI agency are more about. This is related to the discussion about the “absurdity challenge” in the literature. Specifically, some critics have argued that it is absurd to claim that our practices of holding others morally responsible fix the facts of whether one is morally responsible; otherwise, if we hold mentally ill morally responsible, then they are morally responsible, which is absurd (Nelkin 2011; Todd 2016). A recent proposal in response to this challenge suggests differentiating between the criterion thesis and the obtainment thesis (De Mesel & Heyndels 2019), which aligns with what I refer to as the distinction between definition and conditions. Thanks to an anonymous reviewer of this journal for pushing me on this point.

Moreover, Tollon and Tigard argue that AIs meet the exemption conditions, somewhat like children, the mentally ill, and psychopaths. However, the analogy is not unproblematic. Developmental psychology shows that children gradually develop their cognitive and moral capabilities. Psychiatry and clinical psychology reveal that the mentally ill and psychopaths, though lacking specific capabilities, still share key psychological traits with the general adult population. In these cases, we can delineate the boundaries of the shared moral community by comparing these individuals to neurotypical adults. In contrast, we cannot easily compare AIs with normal adult human being in this respect. AIs exceed human performance in many cognitive tasks, such as information gathering and analyzing (e.g., generative AI models like ChatGPT or DeepSeek). Although we struggle to conceive of AIs developing a full psychology profile like humans, excluding them from our moral community requires more substantial justification. The point is, while we find it hard to picture AIs as *recipients* of our reactive attitudes, we might discover new insights by reversing our perspective—namely, by considering them as *expressers* of these attitudes. I will elaborate on this point in the next section.

3 The reversal methodology and the symmetry thesis

Note that one interesting aspect of the discussions in the reactive attitudes framework is that groups typically considered not morally responsible—such as young children, the mentally ill, and individuals with evil upbringings—are viewed as the *recipients* of reactive attitudes. The discussion often focuses on whether these groups are suitable targets for resentment when they cause harm. I suggest shifting the focus from considering them as recipients of reactive attitudes to considering them as *expressers* of these attitudes. This shift can help clarify some of the capacities one needs to qualify as a moral agent in the Strawsonian framework.

Indeed, this idea has been hinted by some philosophers who note that reactive attitudes have corresponding fitting responses in moral “conversations” (Björnsson and Hess 2017; McKenna 2012; Sars 2022). For example, resentment seeks remorse, and it would be meaningless if the target is incapable of remorse. Accordingly, when asking “Are they appropriate targets for resentment?” we can ask, “Are they capable of remorse?” Similarly, some have noted that “anger or resentment are means of eliciting guilt and reparative actions in the offender” (Tigard 2021, p. 603). Accordingly, when asking “Are they appropriate targets for resentment?” we can also ask, “Are they capable of guilt and reparative actions?” In the next section, by drawing on literature of affective computing, I will show the possibility that AIs may be capable of such reactions. For now, let us assume this is

true. In the rest of this section, I will show the plausibility of the reversal methodology by showing that the question of whether one is capable of exchanging reactive attitudes can indeed be answered by examining whether one is capable of *expressing* these attitudes. To do this, I need to argue for The Symmetry Thesis and show that the capacities of *expressing* reactive attitudes are the same as those of *receiving* these attitudes.

According to Strawson, our practice of moral responsibility occurs within a shared moral community where the *exchange* of reactive attitudes takes place. This exchange is *reciprocal* and *mutual*. Therefore, if an individual is exempt from being morally responsible, she is excluded from the shared moral community. Accordingly, an exempt person cannot engage in the reciprocal or mutual exchange of reactive attitudes with others. Thus, when we say that someone is exempt from moral responsibility, this means that this person is not only exempt from *receiving* reactive attitudes but also from *expressing* them. This point is crucial in the Strawsonian framework, given the mutual and reciprocal nature of reactive attitudes.

My reversal methodology relies precisely on this idea. The mutuality and reciprocity of reactive attitudes exchange and human interactions guarantees that a legitimate member of a shared moral community must be capable of both receiving and expressing reactive attitudes. Furthermore, being able to receive and being able to express reactive attitudes require *the very same* capacities, or so I will argue. This Symmetry Thesis is plausible because of the reciprocal nature of reactive attitudes exchange, which is the defining feature of Strawsonian moral responsibility practice. I argue that this is the only plausible way to understand the reciprocal nature of human interaction. To illustrate this, we can use a *reductio ad absurdum* approach. Suppose it is *not the case* that receiving and expressing reactive attitudes require the same capacities. This would mean that one could possess the capacity for one but not the other. Under this assumption, we should be able to imagine the following two types of scenarios:

- a. One is a qualified member of the shared moral community but is always held responsible for their actions and never holds others responsible.
- b. One is a qualified member of the shared moral community but always holds others responsible while never being held responsible themselves.

We can use the concrete example of anger to see how it looks like. The first scenario assumes the possibility that someone can be blamed for their wrongdoing but *either that never cares about others doing wrong or others never commit any wrongdoing*. Similarly, the second scenario assumes that someone always blames others for their wrongdoing but

either always escapes blame for her own wrongdoing or never does anything wrong. All these situations are highly implausible. Therefore, receiving and expressing reactive attitudes must require the very same capacities.⁷

An objection might arise here. One could argue that the capacities for expressing and receiving reactive attitudes are not identical because more is required to be an expresser than a receiver. A clear example is young children. They can express anger as any adult do. However, since they are exempt in the moral community shared by ordinary adult humans, they cannot be the recipients of reactive attitudes. This suggests that the two sets of capacities are not identical.

My response is, because young children are exempt and thus cannot participate in the exchange of reactive attitudes, they cannot, by definition, be recipients or expressers of these attitudes. To understand this, we must carefully distinguish between emotions and reactive attitudes. Consider animal research showing that some animals can also feel anger. Moreover, their anger serves functional interpersonal roles, such as signaling “You should not do that” or “Do not mess with me” (Archer 1988; Panksepp 1998). However, we do not claim that these creatures, capable of expressing real anger, are holding each other morally responsible. In other words, they are neither expressers nor recipients of reactive attitudes in the Strawsonian sense. Similarly, in the case of young children, we might comfortably say, “Young children can feel anger”. But we might feel less comfortable saying, “Young children can feel resentment”. It is the latter rather than the former that is an example of reactive attitudes.⁸ The point is that even if young children are capable of expressing anger, such anger is not a reactive attitude. Therefore, this example does not show that young children are expressers of reactive attitudes and does not refute the thesis that the capacities for expressing and receiving reactive attitudes are the same.

This objection can be framed differently. Some might argue that young children and other exempt groups can have *demands* for goodwill. Such demands make them expressers of reactive attitudes, although they do not qualify to be recipients of these attitudes. My response is that we need to distinguish between *moral agents* and *moral patients* in this context (Regan 1985; Singer 1975). A paradigmatic example of a moral patient is a sentient animal, which can feel pain and thus demands humans refrain from causing them pain.

⁷ One may object that it is logically possible that the capacities to receive and express reactive attitudes are not the same ones, but they are always co-present. I do not deny this possibility. However, even with this weakened thesis, I can still make the case that the capacity of *expressing* reactive attitudes implies the capacity of *exchanging* these attitudes. Nevertheless, it is easier and more plausible to imagine the equal case than the co-present case.

⁸ Although we often loosely talk about anger and resentment interchangeably in the relevant context.

At the same time, sentient animals are not morally responsible for causing pain to humans and are therefore not moral agents. Similarly, although young children have demands for goodwill, these demands merely imply that they are moral patients. Because they are exempt from moral responsibility, they are not (yet) moral agents. An adult human may hold those who have wronged a young child responsible by expressing indignation or resentment on behalf of the child, but the young child herself cannot do this. Therefore, having demands does not imply being a qualified subject capable of expressing reactive attitudes. Thus, the case of the exempt is not a counterexample to the claim that receiving and expressing reactive attitudes require the same capacities.

In summary, the reversal methodology aims to answer the question, “Are they appropriate targets of reactive attitudes?” by answering the question, “Are they appropriate *expressers* of reactive attitudes?” If the answer to the latter question is affirmative, then given that expressing and receiving attitudes engage the same set of capacities—as the Symmetry Thesis requires—we can say that the relevant subjects are capable of exchanging reactive attitudes. Consequently, we can also say that they are morally responsible agents in the Strawsonian sense. The final piece of the puzzle is investigating whether strong AIs are indeed capable of expressing reactive attitudes, which I will argue in the next section.

4 AIs as potential expressers of reactive attitudes

Let me test the limit of the Emotional AI Thesis by investigating how far we can go. This thesis says that future AIs may be capable of expressing reactive attitudes towards humans. I start by showing that they may be capable of certain emotions. To achieve this, we can resort to plenty of resources in the philosophy and psychology of emotions. Although debates about how to define emotions continue, many scholars generally agree that emotions involve aspects of cognition, physiology (including brain activities), experience, and motivation (Dukes et al. 2021). In fields like psychology, philosophy, and affective neuroscience, which have extensively studied emotions, researchers propose that emotions are tools that humans use to appraise or evaluate real-life scenarios and react to them with a series of responses (Frijda 1986; LeDoux 1998; Prinz 2004; Roberts 2003; Scarantino 2014; Scherer 1982; Tappolet 2016). For example, anger occurs when the subject evaluates something as offensive or obstructive to her goal. The angry subject is motivated to take actions to remove the obstacle or source of offense. Across this research, emotions are seen as serving specific functions that help people solve problems.

The field of affective computing, which intersects both emotions and AIs, has developed similar ideas. This field

studies how to measure emotion and “use the knowledge to engineer smarter technology and to create experiences that improve lives”, among other goals (Picard 2015, p. 49). Affective computing has primarily relied on the appraisal model of emotions, developed over generations by psychologists (Arnold 1960; Ellsworth and Scherer 2003; Lazarus 1991; Ortony et al. 1988; Scherer 2001). While appraisal theories vary in detail, they generally consider emotions as the subject’s appraisal of the environment with respect to her well-being. The appraisal can have several dimensions, including valence (pleasure or displeasure), arousal (intensity of the state), certainty (the degree of certainty that the object poses), control or potency (whether the subject feels powerful to handle the object), coping strategy (ways to deal with the object), etc. The purpose of forming such appraisal states is to fulfill the subject’s goal of survival and well-being. For example, when encountering a bear in the wild, fear may arise from appraising the bear as highly threatening, the situation as highly unpleasant, feeling a lack of control, and deciding that fleeing is the best strategy for survival.

Appraisal theories do not require specific physiological changes unique to humans to define an emotion. This flexibility allows affective computing theorists to apply these theories to create emotion-based agent architecture.⁹ These agents are most prominently developed in the field of Human–Computer Interaction (HCI). In the HCI context, humans are regarded as users, and agents are primarily designed to adapt to users’ emotions to enhance the interaction experience. This involves designing *sensors* that detect the user’s emotional states through facial expressions, gestures, vocal intonation (such as volume and pitch), sensorimotor cues (such as pressure), natural language, and autonomic nervous system signals (such as skin conductance rates, heart rate, blood pressure, and respiration rates). From the perspective of the computer agents, this is the receiving process. But interaction requires both receiving and reacting. Accordingly, HCI agents are also equipped with *actuators* that control anthropomorphic embodiments capable of exhibiting the measures mentioned above, allowing for the transmission of emotional signals back to humans. Thus, computer agents can not only interpret emotional signals from humans but also send corresponding emotional signals back to them.

It is often assumed that such computer agents are limitedly emotionally toned, primarily serving human needs and pleasure, avoiding so-called negative emotions like anger,

fear, sadness, or disgust, emotions more related to the context of moral responsibility. However, this does not mean that they are restricted to only pleasant emotions. There is no evidence suggesting that positive and negative emotions, as *two categories*, require distinct capacities—whether in the appraisal process or in physiological or neurological structure—even though different discrete emotions may involve different faculties. Therefore, if necessary, a computer agent could plausibly be designed to display a full spectrum of emotions, aligning more closely with the concept of strong AIs.

We must proceed cautiously, however. It remains unclear whether the emotions displayed by current emotional computer agents are genuine emotions or mere reflexes. For example, a standard computer agent operates in three steps: “(1) sensing a set of triggers (e.g., dangers, appeals) specific to its physicality or embodiment; (2) responding to these signals with action tendencies (approach, avoid, attend, reject, interrupt) implemented as a reflex-based agent using action-reaction rules; and (3) actuating these actions via its actuators...in a manner that is psychologically valid” (Lisetti and Hudlicka 2015, p. 167). This process mirrors how human emotions function, as formulated by many emotion scholars: emotions detect and motivate. However, in the second step, the action tendencies are based on action-reaction rules resembling reflex-like connections. These rules are largely “if–then” rules, such as “if car-in-front-is-braking, then initiate-braking” (Russell and Norvig 2010, p. 50). It is difficult to determine whether all reactions based on these rules are analogous to human emotional responses, as it is unclear if human emotional responses can be reduced to a set of complex rules like this. Perhaps only more advanced affective neuroscience can provide an answer. It is also premature to judge whether this rule-based mechanism would be compatible with appraisal, but there is no reason to dismiss this possibility at this point.

To clarify this last point, the “if–then” rule should be distinguished from the “if→do” rule. Some scholars argue that this distinction is crucial between humans and non-human animals. It has been suggested that non-human animals act according to the “if→do” rule, which essentially follows a stimulus–response mechanism. However, this is not causal reasoning, which is typical in human reasoning process. The distinction lies in the fact that causal reasoning allows for hypothetical or counterfactual reasoning, which does not necessarily lead to actions but rather entertains different possibilities. This process is seen as a symbol of reflection or deliberation. Importantly, the “if–then” rule that AIs follow is not incompatible with human causal and hypothetical reasoning. It has been proposed that we can implement a form of causal reasoning in computers (Pearl 2009).

The upshot is that future AIs may develop emotions, understood functionally. Next, I need to show that they can

⁹ In this context, a computer “agent” is different from a human “agent” and a moral “agent”. A computer agent is “something that perceives and acts in an environment” (Russell and Norvig 2021, 60). I hope the context where the two senses are used respectively in this paper are obvious enough to avoid any confusion.

express reactive attitudes, which are a particular subset of emotions.¹⁰ I will argue for this by replying to two crucial series of objections.

4.1 Objection one

The first objection is: even if AIs can display emotions like fear, this may not be sufficient for them to express emotions categorized as reactive attitudes. As discussed earlier, these emotional attitudes include resentment, blame, guilt, gratitude, forgiveness, and so on. These emotions appear to be highly cognitive. It is these emotions that is the basis of our practice of moral responsibility attribution.

My response is that while this concern may be significant when discussing whether non-human animals (a usually juxtaposed topic) or young children can be considered moral agents, it is less relevant for AIs. Scholars often refer to “highly cognitive emotions” partly to preserve certain types of emotions unique to humans. These emotions are social and require significant cognitive capacities such as conceptual capabilities, a sense of self and others, perspective-taking, and more. Even though non-human animals might experience fear or anger (more commonly referred to as “aggression”), they are not considered moral agents due to their lack of these more advanced capacities. However, these considerations are less pertinent when discussing AIs, which are renowned for their intelligence, exemplifying cognitive capacities.

One may further object that even if AIs possess intelligence, this does not necessarily mean that they have the cognitive capacities relevant to having reactive attitudes. In response, I refer to observations by affective computing scientists Christine Lisetti and Eva Hudlicka: “[t]he field of artificial intelligence, which formally emerged in 1956, founded most of its models of intelligence on previously established affectless theories of intelligence, originally rooted exclusively in logic” (Lisetti and Hudlicka 2015, 160). This implies the possibility of a new understanding of “intelligence”. After all, the concept of “emotion intelligence”, which was introduced by Dan Goleman and refers to the capability to feel the right emotion at the right time, is already widely accepted (Goleman 1996). Additionally, Rosalind Picard, a pioneer in affective computing, noted that she began to pay attention to emotions in the then orthodox research in computer science because she noticed the

significance of emotions in human intelligence (Picard 2015).¹¹ Thus, there is no preemptive reason to rule out the possibility that cognitive capacities related to emotions in general and to reactive attitudes in particular can be modeled in strong AIs. That is precisely what researchers in affective computing are exploring.

4.2 Objection two

Another concern is the role of consciousness in the possibility of artificial emotions. As mentioned earlier, the standard view of emotions in current research takes emotions to be qualitatively felt states. Similarly, in folk psychology, the experiential feature of emotions is often considered their most definitive feature. This is why people commonly use “emotion” and “feeling” interchangeably. If we accept this view, does it not mean that I must consider the consciousness criterion?

My response is two-fold. First, it is not necessarily bad news if we have to address the question of whether future AIs can possess consciousness. Even though David Chalmers has famously proposed the “hard problem” of consciousness, he and some other philosophers who also appreciate this problem do not dismiss the possibility that consciousness could arise in AIs. In fact, Chalmers himself supports the view that AIs can have consciousness based on a computational framework of the mind (Chalmers 2011). In this framework, a computation and a physical system are linked through the process of implementation, so that the physical system realizes the computation and the computation describes the physical system. According to Chalmers, an implementation is successful when the causal structure of the physical system mirrors the formal structure of the computation. Besides, he argues that cognitive systems have mental properties in virtue of their causal organizations. Thus, the mind, or our cognitive systems, can have a computational interpretation. The crucial bridge in the theory is the concept of causal organization. Chalmers argues that mental states are organizational invariant, meaning that they do not change the causal topology of the cognitive processes. This allows for the possibility that two systems made

¹⁰ Some may object that not all reactive attitudes are emotions. For example, there is non-emotional theory of blame (Hieronimi 2004). However, scholars have argued that this theory is implausible because it risks conflating blaming with judging blameworthy (Coates and Tognazzini 2012). I find this criticism plausible. Besides, this paper proceeds within the Strawsonian reactive attitudes framework, which is committed to an emotional account of blaming.

¹¹ Here is what she says in the article, “[w]e came up with many clever ways to parallelize, pipeline, optimize, and otherwise process sounds and sights and other signals humans usually interpret effortlessly. However, never did anyone figure out how to give a computer anything like motivation, drive, and a sense of how to evaluate shifting priorities in a way that acted genuinely intelligent” (Picard 2015, 41). To be fair, emotions are not entirely left out of the picture in the research of human non-irrational cognitions (I said “non-irrational” because there is a long history in the Western tradition to associate irrationality with emotion). Herbert Simon has acknowledged emotions as “interruptive systems” in his seminal paper in 1967 (Simon 1967). However, it is unclear to what extent this proposal has been implemented in designing artificial intelligence.

of different materials, such as neurons and silicon, can both possess mental states because mental states, as organizational invariants, do not change the causal topology of the systems. These mental states include not only psychological functional states, such as beliefs, but also phenomenal states, such as qualia. Admittedly, Chalmers' theory of organizational invariants and his arguments for mental states as organizational invariants can be debated. However, the idea is not evidently implausible. Since the debate remains unresolved, the possibility of conscious AIs still exists.

Second, despite that the question about consciousness is also relevant to the current discussion, it becomes more concrete when approached from the perspective of reactive attitudes. That is why the reactive attitudes approach is particularly helpful. Within this framework, the key question is whether a being can have certain emotions. Some scholars have argued that functionally sophisticated AIs may have emotions and be appropriate objects of reactive attitudes because phenomenal consciousness is not essential to emotions (Antill 2024; Vaidya 2024). I also emphasize the functional role as the essential feature of emotions and reactive attitudes. If a certain kind of evolutionary theory is correct, humans possess emotions that serve functions, including the interpersonal function of communication. However, there is no reason to assume that this function can only be realized through *conscious* emotions. It is conceivable that some beings can have attitudes that fulfill these relevant functions without having phenomenally conscious feelings. Perhaps this function is carried out in a subconscious process, with the end result being accessed to the subject. Expressing and exchanging reactive attitudes without consciousness does not imply that the subject is unaware of these exchanges. In other words, even if AIs cannot be phenomenally conscious, as long as they can exchange their version of reactive attitudes (emotion without the accompanying feeling), they can be regarded as members of the shared moral community.

Gunnar Björnsson and Kendy Hess have argued that collective agents, such as corporations, can have the functional equivalents of beliefs, desires, intentions, and emotions without the phenomenological feature (Björnsson and Hess 2017). The idea is similar in AIs. One objection to this reasoning is that *sincere* emotions, which are required in these discussions, require the exact phenomenological feature (Shoemaker 2019). For example, if one does not feel the pain involved in resentment, one cannot be in sincere resentment. While this presents a challenge to the case of collective responsibility, it is less threatening to strong AIs. Consider Samantha from the movie *Her*, or Adam from the novel *Machines Like Me*. As a computer operational system, Samantha can effectively exchange emotions with the human protagonist. Similarly, as a human-shaped robot, Adam can also effectively exchange emotions with its "owners". My intuition is that they are capable of sincere emotions.

Accordingly, if we insist that they are only capable of functional equivalents of emotions without the phenomenological feature, then these functional equivalents are sufficient to qualify them as members of the shared moral community. If we find it more natural to attribute phenomenological features to characters like Samantha or Adam, then we will be already acknowledging their capacity for consciousness.

To sum these up, within the reactive attitudes framework, the essential aspect is the exchange of reactive attitudes in a shared moral community. Although the capacities of members of this community are important (as they determine whether the reactive attitudes are appropriate or should be withheld), they are significant because of the practice of exchanging reactive attitudes. Because future AIs may be capable of expressing and thus exchanging reactive attitudes within a moral community shared with humans, they may qualify as morally responsible agents. Despite the fact that our sensibilities may be modified because of the inclusion of AIs (Sars 2022), this "organic development" (Shoemaker 2017) does not undermine the reactive attitudes exchange system that we share with each other.

5 Conclusion

In this paper, I applied Strawsonian reactive attitude theories of moral responsibility to artificial intelligence to argue that AIs may be morally responsible agents. After explaining the Strawsonian framework, I showed how this framework advances traditional discussions on intentionality and consciousness, thereby enhancing our understanding of AI moral agency. Within a plausible functional account of emotions, we saw that AIs may be held morally responsible because they may effectively and aptly express and accept reactive attitudes, which establishes their status as members of the shared moral community with humans. In comparison to the approach that investigates whether AIs are appropriate *recipients* of reactive attitudes, the approach I developed here focuses on whether AIs can be apt *expressers* of these attitudes. This shift in perspective can provide new resources for addressing questions such as whether AIs can have a quality of will, can care, and can justify their reasons.

The question about whether AIs can be morally responsible is important. It not only addresses how we should interact with certain beings (in this case, AIs), which is already a challenging issue, but also helps us better understand our practice of moral responsibility attribution in daily life. Notably, one major advantage of the Strawsonian framework is that it avoids over-intellectualizing moral responsibility by sidestepping the free will and determinism debate. In this sense, investigating AI and moral agency also demonstrates the reflective equilibrium methodology, where one focuses on the practical issue of AI responsibility while

simultaneously considering the reactive attitude theories. This discussion is not only about the implications of the reactive attitudes approach for AIs; it also invites a reevaluation of the reactive attitudes approach itself. Some may view AIs as counterexamples to the Strawsonian framework of moral responsibility, arguing that while AIs may be able to exchange reactive attitudes, they are not moral agents. However, this would require a robust account of moral responsibility at the first place.

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References

- Antill G (2024) Robots and reactive attitudes: a defense of the moral and interpersonal status of non-conscious agents. *AI Ethics* 5(2):1651–1667
- Archer J (1988) *The behavioural biology of aggression*, vol 1. Cup Archive
- Arnold MB (1960) *Emotion and personality*
- Begin D (2018) Responsibility, libertarians, and the “facts as we know them”: a concern-based construal of Strawson's reversal. *Ethics* 128(3):612–625
- Björnsson G, Hess K (2017) Corporate crocodile tears? *Philos Phenomenol Res* 94(2):273–298
- Chalmers D (1997) *The conscious mind: in search of a fundamental theory*. Oxford Paperbacks
- Chalmers D (2011) A computational foundation for the study of cognition. *J Cogn Sci* 12(4):323–357
- Coates DJ, Tognazzini NA (2012) The nature and ethics of blame. *Philos Compass* 7(3):197–207
- Coeckelbergh M (2010) Moral appearances: emotions, robots, and human morality. *Ethics Inf Technol* 12:235–241
- Coeckelbergh M (2020) Artificial intelligence, responsibility attribution, and a relational justification of explainability. *Sci Eng Ethics* 26(4):2051–2068
- Danaher J (2016) Robots, law and the retribution gap. *Ethics Inf Technol* 18(4):299–309
- Darwall S (2006) *The second person standpoint: morality, respect, and accountability*. Cambridge press
- De Mesel B, Heyndels S (2019) The facts and practices of moral responsibility. *Pac Philos Q* 100(3):790–811
- Dukes D, Abrams K, Adolphs R, Ahmed ME, Beatty A, Berridge KC, Broomhall S, Brosch T, Campos JJ, Clément F, Cunningham WA, Damasio A, Damasio H, D'Arms J, Davidson JW, de Gelr B, Deonna J, de Sousa R, Ekman P, Ellsworth PC, Fehr E, Fischer A, Foolen A, Frevert U, Grandjean D, Gratch J, Greenberg L, Greenspan P, Gross JJ, Halperin E, Kappas A, Keltner D, Knutson B, Konstan D, Kret ME, LeDoux JE, Lerner JS, Levenson RW, Loewenstein G, Manstead ASR, Maroney TA, Moors A, Niedenthal P, Parkinson B, Pavlidis I, Pelachaud C, Pollak SD, Pourtois G, Roettger-Roessler B, Russell JA, Sauter D, Scarantino A, Scherer KR, Stearns P, Stets JE, Tappolet C, Teroni F, Tsai J, Turner J, Reekum CV, Vuilleumier P, Wharton T, Sander D, Clay Z (2021) The rise of affectivism. *Nat Hum Behav* 5(7):816–820
- Ellsworth PC, Scherer KR (2003) *Appraisal processes in emotion*. Oxford University Press
- Fischer JM, Ravizza M (1998) *Responsibility and control: a theory of moral responsibility*. Cambridge University Press
- Floridi L, Sanders JW (2004) On the morality of artificial agents. *Minds Mach* 14:349–379
- Frijda NH (1986) *The emotions*. Cambridge University Press
- Goleman D (1996) Emotional intelligence. Why it can matter more than IQ. *Learning* 24(6):49–50
- Hieronymi P (2004) The force and fairness of blame. *Philos Perspect* 18:115–148
- Hieronymi P (2020) *Freedom, resentment, and the metaphysics of moral*, vol 50. Princeton University Press
- Himma KE (2009) Artificial agency, consciousness, and the criteria for moral agency: what properties must an artificial agent have to be a moral agent? *Ethics Inf Technol* 11:19–29
- Himmelreich J (2019) Responsibility for killer robots. *Ethical Theory Moral Pract* 22(3):731–747
- Johnson DG (2006) Computer systems: moral entities but not moral agents. *Ethics Inf Technol* 8:195–204
- Kraaijeveld SR (2020) Debunking (the) retribution (gap). *Sci Eng Ethics* 26(3):1315–1328
- Lazarus RS (1991) *Emotion and adaptation*. Oxford University Press
- LeDoux JE (1998) *The emotional brain: the mysterious underpinnings of emotional life*. Simon and Schuster
- Levy N (2003) Cultural membership and moral responsibility. *Monist* 86(2):145–163
- Levy N (2007) The responsibility of the psychopath revisited. *Philos Psychiatry Psychol* 14(2):129–138
- Lisetti C, Hudlicka E (2015) Why and how to build emotion-based agent architectures. *The Oxford handbook of affective computing*, pp 94–109
- Matthias A (2004) The responsibility gap: ascribing responsibility for the actions of learning automata. *Ethics Inf Technol* 6:175–183
- McKenna M (2012) *Conversation and responsibility*. Oup Usa
- Nelkin DK (2011) *Making sense of freedom and responsibility*. OUP Oxford
- Oimann AK, Tollon F (2024) Responsibility gaps and technology: old wine in new bottles? *J Appl Philos*. <https://doi.org/10.1111/japp.12763>
- Ortony A, Clore GL, Collins A (1988) *The Cognitive structure of emotions*. Cambridge. Cambridge University Press
- Panksepp J (1998) *Affective neuroscience*. Oxford University Press
- Pearl J (2009) *Causality*. Cambridge University Press
- Picard RW (2015) The promise of affective computing. *The Oxford handbook of affective computing*, pp 11–20
- Prinz J (2004) *Gut reactions: a perceptual theory of emotion*. Oxford university Press
- Rebera AP (2024) Reactive attitudes and AI-agents—making sense of responsibility and control gaps. *Philos Technol* 37(4):126

- Regan T (1985) The case for animal rights. University of California Press, Berkeley, CA
- Roberts RC (2003) Emotions: an essay in aid of moral psychology. Cambridge University Press
- Russell P (2004) Responsibility and the condition of moral sense. *Philos Topics* 32(1/2):287–305
- Russell A, Norvig P (2010) Artificial intelligence a modern approach. Pearson Education Inc
- Sars N (2022) Engineering responsibility. *Ethics Inf Technol* 24(3):32
- Scarantino A (2014) The motivational theory of emotions
- Scherer KR (1982) The nature and function of emotion. Sage Publications
- Scherer KR (2001) Appraisal considered as a process of multilevel sequential checking. *Apprais Process Emotion Theory Methods Res* 92(120):57
- Searle JR (1992) The rediscovery of the mind. MIT press
- Sebo J, Long R (2023) Moral consideration for AI systems by 2030. *AI Ethics*. <https://doi.org/10.1007/s43681-023-00379-1>
- Shoemaker D (2015) Responsibility from the Margins. Oxford University Press
- Shoemaker D (2017) Response-dependent responsibility; or, a funny thing happened on the way to blame. *Philos Rev* 126(4):481–527
- Shoemaker D (2019) Blameworthy but unblamable: a paradox of corporate responsibility. *Geo JL Pub Pol'y* 17:897
- Shoemaker D, Nelkin D, Pereboom D (2022) Response-dependent theories of responsibility. *The Oxford Handbook of Moral Responsibility*, pp 304–323
- Simon HA (1967) Motivational and emotional controls of cognition. *Psychol Rev* 74(1):29
- Singer P (1975) Animal liberation. Harper Collins, New York
- Smith N, Vickers D (2021) Statistically responsible artificial intelligences. *Ethics Inf Technol* 23(3):483–493
- Sparrow R (2007) Killer robots. *J Appl Philos* 24(1):62–77
- Strawson PF (2008) Freedom and resentment and other essays. Routledge
- Talbert M (2012) Moral competence, moral blame, and protest. *J Ethics* 16:89–109
- Tappolet C (2016) Emotions, values, and agency. Oxford University Press
- Tigard DW (2021) There is no techno-responsibility gap. *Philos Technol* 34(3):589–607
- Todd P (2016) Strawson, moral responsibility, and the “order of explanation”: an intervention. *Ethics* 127(1):208–240
- Tollefsen D (2003) Participant reactive attitudes and collective responsibility. *Philos Explor* 6(3):218–234
- Tollon F (2023) Responsibility gaps and the reactive attitudes. *AI Ethics* 3(1):295–302
- Vaidya AJ (2024) Can machines have emotions? *AI Soc*. <https://doi.org/10.1007/s00146-024-02022-x>
- Wallace RJ (1994) Responsibility and the moral sentiments. Harvard University Press
- Watson G (1987) Responsibility and the limits of evil: variations on a Strawsonian theme. *The philosophy of free will: essential readings from the contemporary debate*, pp 84–113
- Wolf S (1987) Sanity and the metaphysics of responsibility. *Ethical theory: an anthology*

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